

Vel Tech Group of Institutions

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Batch: 2028

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2028_NeoColab_Competitive Coding II_L8

VTU_Week 6_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Yuvana, an avid programmer, is keen on mastering various sorting algorithms and has recently delved into Counting Sort. As a practice exercise, she is given an array of N integers where each element ranges from 1 to $N-1$. Some numbers may appear multiple times in the array. Her task is to identify and print all numbers that appear more than once. If no such duplicates are found, she should print -1.

Help Yuvana by implementing a function that efficiently finds and prints duplicate numbers using the Counting Sort technique.

Input Format

The first line of input consists of an integer N , representing the number of elements in the array.

The second line of input consists of N space-separated integers, representing the elements of the array.

Output Format

The output prints all the duplicate numbers that appear more than once in the array, each separated by a space.

If no duplicates are found, print -1.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

1 3 4 2 2

Output: 2

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int arr[20];
```

```
    int count[21] = {0}; // max size
```

```
    // input + count
```

```
    for (int i = 0; i < n; i++) {
```

```
        scanf("%d", &arr[i]);
```

```
        count[arr[i]]++;
```

```
    }
```

```
    int found = 0;
```

```
    // print duplicates
```

```
    for (int i = 1; i <= n; i++) {
```

```
        if (count[i] > 1) {
```

```
        printf("%d ", i);  
        found = 1;  
    }  
}  
  
if (!found) {  
    printf("-1");  
}  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Anu is tasked with organizing a list of fruit names stored as strings. To ensure that the list is easy to navigate, she decides to sort the strings in ascending alphabetical order.

Being a fan of efficient sorting algorithms, Anu opts to use radix sort for this task. Help Anu with a program that can take a list of fruit names as input and output the sorted list.

Input Format

The first line of input consists of an integer n , representing the number of fruit names in the list.

The next n lines each consist of a string, representing the names of the fruits.

Output Format

The output prints n lines containing the sorted list of fruit names in ascending alphabetical order.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

apple

banana

orange

grape

Output: apple

banana

grape

orange

Answer

// You are using GCC

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    char arr[20][100];
```

```
    // input strings
```

```
    for (int i = 0; i < n; i++) {
```

```
        scanf("%s", arr[i]);
```

```
    }
```

```
    // sort strings (ascending)
```

```
    for (int i = 0; i < n - 1; i++) {
```

```
        for (int j = i + 1; j < n; j++) {
```

```
            if (strcmp(arr[i], arr[j]) > 0) {
```

```
                char temp[100];
```

```
                strcpy(temp, arr[i]);
```

```
                strcpy(arr[i], arr[j]);
```

```
                strcpy(arr[j], temp);
```

```
            }
```

```
        }
```

```
    }
```

```
    // print sorted list
```

```
    for (int i = 0; i < n; i++) {
```

```
        printf("%s\n", arr[i]);
```

```
    }
```

```
} return 0;
```

Status : Correct

Marks : 10/10

3. Problem Statement

Riya, an HR manager, needs a quick way to identify the Kth highest salary among employees for bonus distribution. To efficiently retrieve this information, she wants to use the max-heap technique.

Help Riya implement a solution that finds the Kth highest salary from the given list of salaries.

Input Format

The first line of input consists of an integer N, representing the number of employees.

The second line consists of N space-separated integers, representing the salaries of N employees.

The third line consists of an integer K, indicating the rank of the desired Kth highest salary.

Output Format

The output prints: <k>th largest Salary: <result>

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 7
30000 25000 40000 35000 20000 28000 29500
5

Output: 5th largest Salary: 28000

Answer

```

// You are using GCC
#include <stdio.h>

int main() {
    int n, k;
    scanf("%d", &n);

    int arr[105];

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    scanf("%d", &k);

    // sort in descending order
    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (arr[i] < arr[j]) {
                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
    }

    printf("%dth largest Salary: %d", k, arr[k - 1]);

    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Manage a warehouse inventory with unique barcode numbers utilizing a specialized sorting algorithm designed for barcodes with exactly 16 digits. Input the number of barcode numbers (N) and their corresponding values.

Employ the Bucket Sort algorithm to distribute and sort the numbers based

on their digits. Display the sorted barcode numbers, ensuring an efficient and organized warehouse inventory management system tailored for barcode systems with 16 digits.

Input Format

The first line contains an integer N, representing the number of barcode numbers to be sorted.

The following N lines contain exactly 16-digit long integers, each representing a unique barcode number.

Output Format

The output displays a single line containing "Sorted barcode numbers:" followed by the sorted barcode numbers in ascending order. Each barcode number is displayed on a separate line.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

2345678901234567

1234567890123456

3456789012345678

5678901234567890

4567890123456789

Output: Sorted barcode numbers:

1234567890123456

2345678901234567

3456789012345678

4567890123456789

5678901234567890

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
scanf("%d", &n);  
long long arr[100];  
  
// input  
for (int i = 0; i < n; i++) {  
    scanf("%lld", &arr[i]);  
}  
  
// sort ascending  
for (int i = 0; i < n - 1; i++) {  
    for (int j = i + 1; j < n; j++) {  
        if (arr[i] > arr[j]) {  
            long long temp = arr[i];  
            arr[i] = arr[j];  
            arr[j] = temp;  
        }  
    }  
}  
  
// output  
printf("Sorted barcode numbers:\n");  
for (int i = 0; i < n; i++) {  
    printf("%lld\n", arr[i]);  
}  
  
return 0;  
}
```

Status : Correct

Marks : 10/10